

The empirical recurrence for OEIS A283548: recovering a conjecture that is not written down

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Abstract

OEIS A283548 records “Empirical recurrence of order 69”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 121$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 69, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 69 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283548 is “Number of $7 \times n$ 0..1 arrays with no 1 equal to more than one of its horizontal, diagonal and antidiagonal neighbors.”. It has offset 1 and begins

128, 2570, 54211, 1529463, 34266804, 854841910, 20707472573, 500387112083, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Mar 10 2017 09:35:39, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2698$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 121$ of the 2698, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 121$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 242$ exact terms of the model returns order 69 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	9	c_{19}	−125328564872646	c_{37}	−187782320933737694402	c_{55}	171652685907314
c_2	247	c_{20}	−545727878110230	c_{38}	−58179331540661841753	c_{56}	−23479192574054
c_3	3001	c_{21}	415746828225028	c_{39}	24037867230986631444	c_{57}	−57331849534431
c_4	−3313	c_{22}	2551784103302592	c_{40}	−235700099836157851196	c_{58}	137981146860320
c_5	98625	c_{23}	10628060895894894	c_{41}	563263729804180468806	c_{59}	19742758140049
c_6	−486138	c_{24}	14250485846999779	c_{42}	1564734006975964870110	c_{60}	−12354003357784
c_7	5118951	c_{25}	22238304727488749	c_{43}	307281396446683912799	c_{61}	−1234869605220
c_8	−82735042	c_{26}	−31281858813486799	c_{44}	−2513586928180200744759	c_{62}	27614899378939
c_9	580298085	c_{27}	−144807838745526112	c_{45}	−1368984584829801519854	c_{63}	−7167389215223
c_{10}	−4516263538	c_{28}	−311540099647655065	c_{46}	2574582765508267238153	c_{64}	−462529166894
c_{11}	24552971517	c_{29}	12895518193551707	c_{47}	1314106340508646164763	c_{65}	873692116445
c_{12}	−120009209226	c_{30}	1104520399991560872	c_{48}	−3332566457752048273589	c_{66}	89736905495
c_{13}	532518338812	c_{31}	1184472809433177129	c_{49}	−2321165498173480204315	c_{67}	16862828807
c_{14}	−1572537892600	c_{32}	−633959127522156021	c_{50}	2447684238143483480816	c_{68}	1012524221
c_{15}	4631333832093	c_{33}	2330841979782214310	c_{51}	5201154367781403606255	c_{69}	178870272
c_{16}	−5464487108352	c_{34}	−13652563502345465902	c_{52}	748530129755161048322		
c_{17}	−897423918848	c_{35}	−67445097861606977650	c_{53}	−1427681943521061879180		
c_{18}	−14745239669768	c_{36}	−132632023864888077677	c_{54}	483960847849544545818		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 69, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $69 + 4$ terms removed returns order 69 as well, so $\deg q_{\min} = 69 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $69 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 69 that this sequence satisfies — and that family turns out to have exactly one member.

References

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